

DS/ML Roadmap vs AI Engineer Roadmap - Comparison

Overview

DS/ML Roadmap: Focuses on Data Science, Machine Learning, Deep Learning, and predictive modeling.

AI Engineer Roadmap: Focuses on LLMs, Prompt Engineering, RAG, AI Agents, and production AI systems.

Key Comparison

- Goal: Predictive Models vs AI Applications
- Week 1: EDA & Data Wrangling vs LLM Foundations & Prompt Engineering
- Week 2: ML Algorithms vs LangChain, Memory & Vector Databases
- Week 3: Deep Learning & Transfer Learning vs RAG & AI Agents
- Week 4: ML Deployment vs AI Product Deployment with FastAPI & Docker

Projects

DS/ML:

1. Exploratory Data Analysis Report
2. Churn/Loan Prediction Model
3. Image Classifier
4. End-to-End ML Application

AI Engineer:

1. NL-to-SQL Assistant
2. Data Analysis Chatbot
3. AI Research Agent
4. Business Intelligence AI Assistant

Skills After Completion

DS/ML: Pandas, Scikit-Learn, XGBoost, PyTorch, Keras, Model Evaluation.

AI Engineer: Prompt Engineering, OpenAI APIs, LangChain, LangGraph, RAG, Agents, FastAPI, Docker.

Recommendation for Twinkle's Goals

Since the goal is to master Python, Prompt Engineering, and AI, a blended approach is ideal:

70% AI Engineer Roadmap + 30% DS/ML Roadmap.

Keep ML fundamentals, EDA, XGBoost, and model evaluation from DS/ML while prioritizing LLMs, RAG, Agents, and AI deployment.